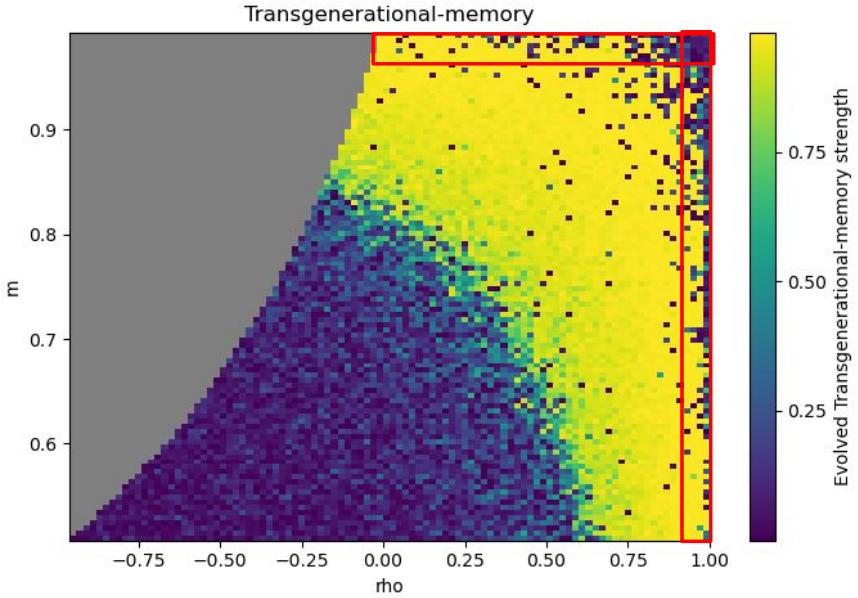
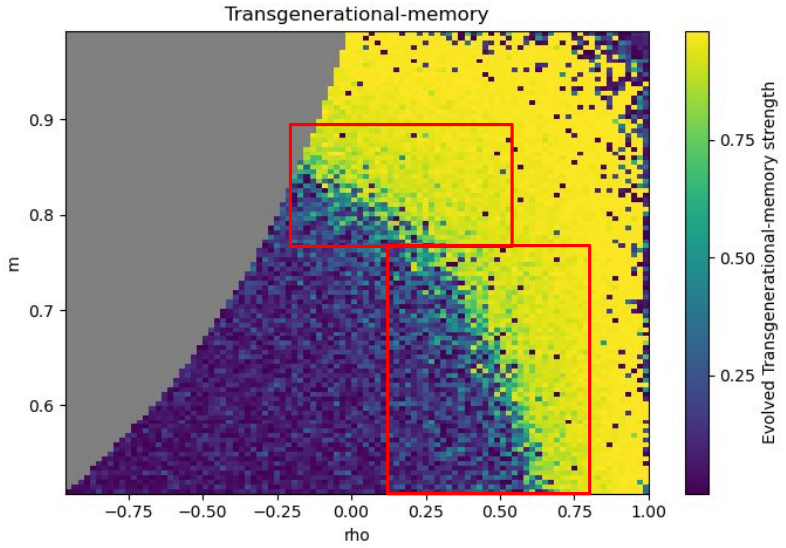
Question 1: what does increasing dimension do to transgenerational memory

I want to show that the number of ‘worlds’ where memory evolves as well as the value of the evolved memory increases with increasing dimensions

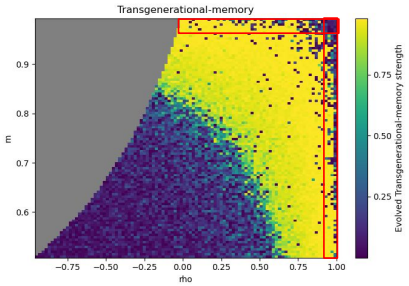
Strategy 1: Count all the worlds in different portions of the heatmap that are > 0.75 (high) and <0.25 (low) memory



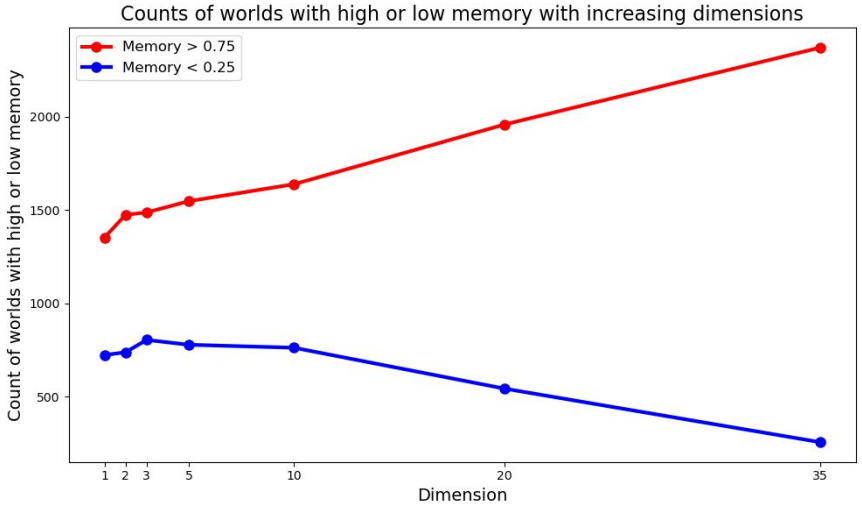
Zone 1: right corner

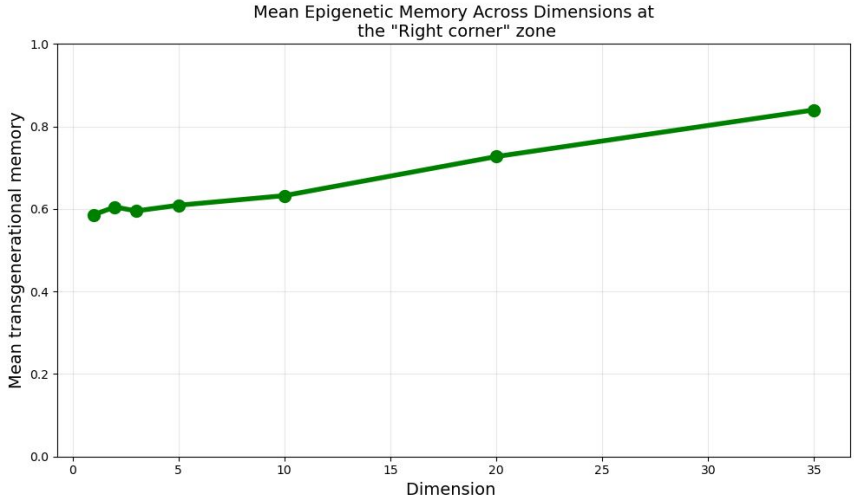


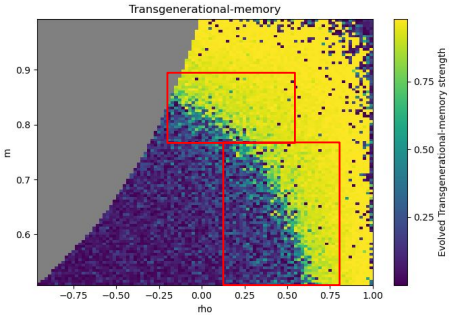
Zone 2: Center strip



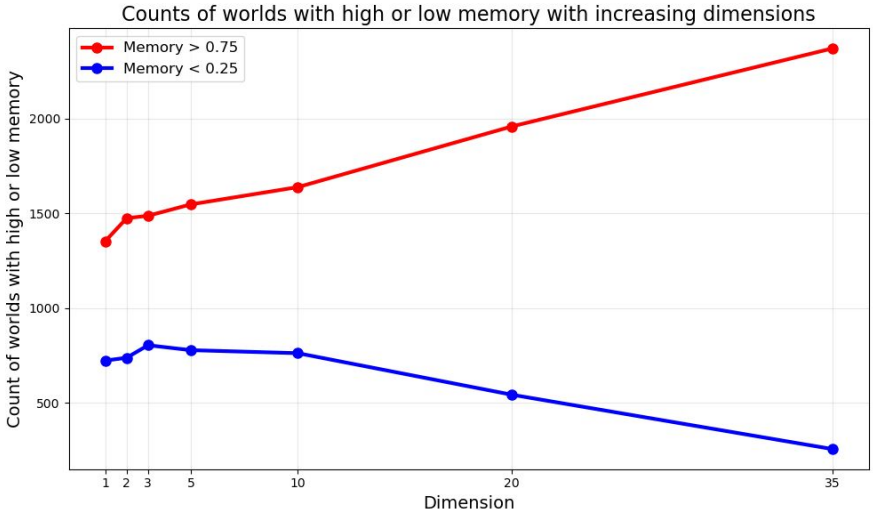
Zone 1: right corner

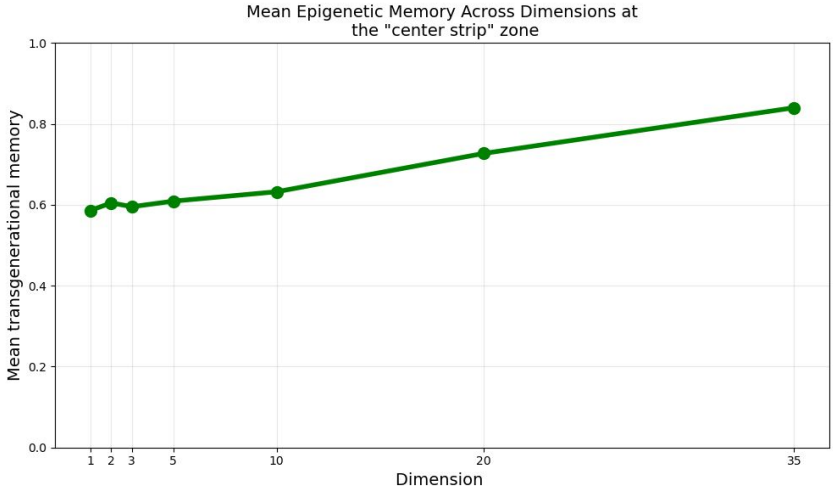




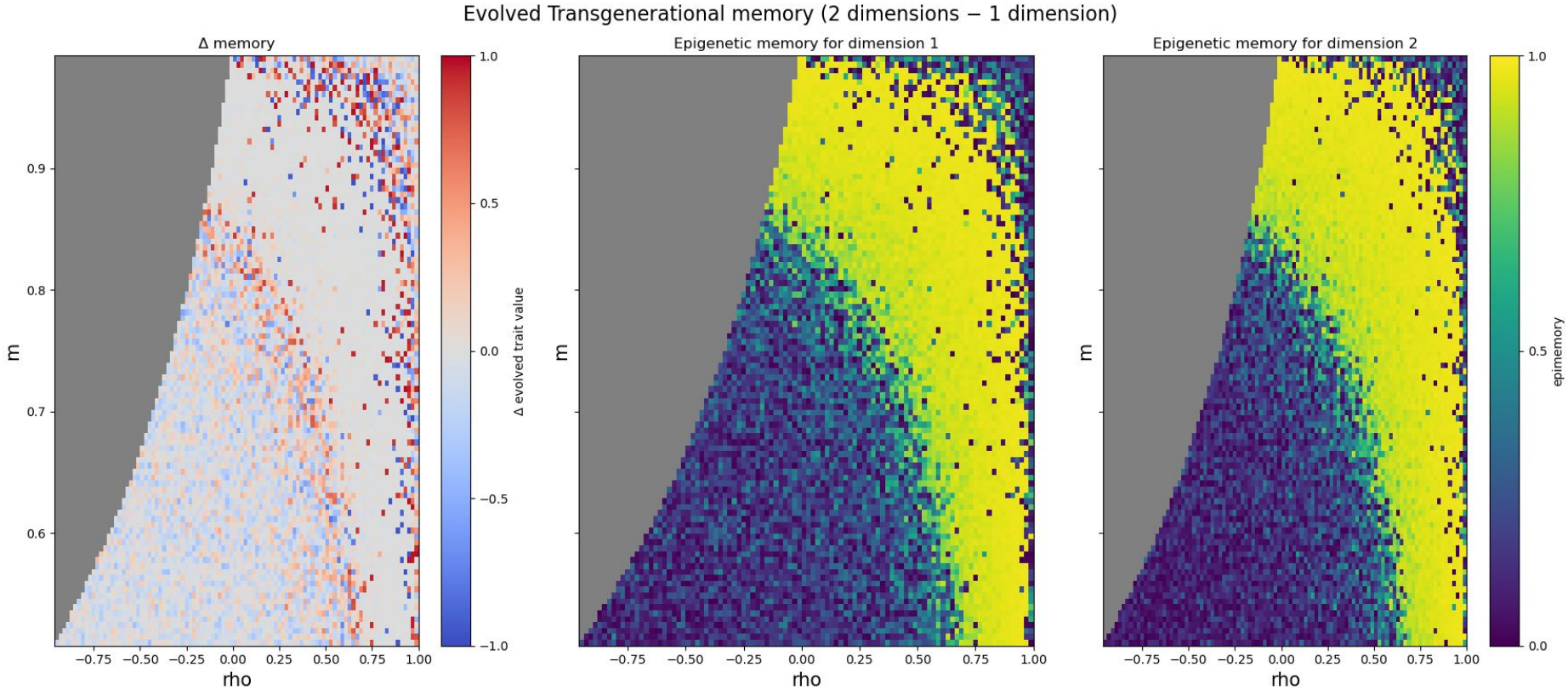


Zone 2: Center strip



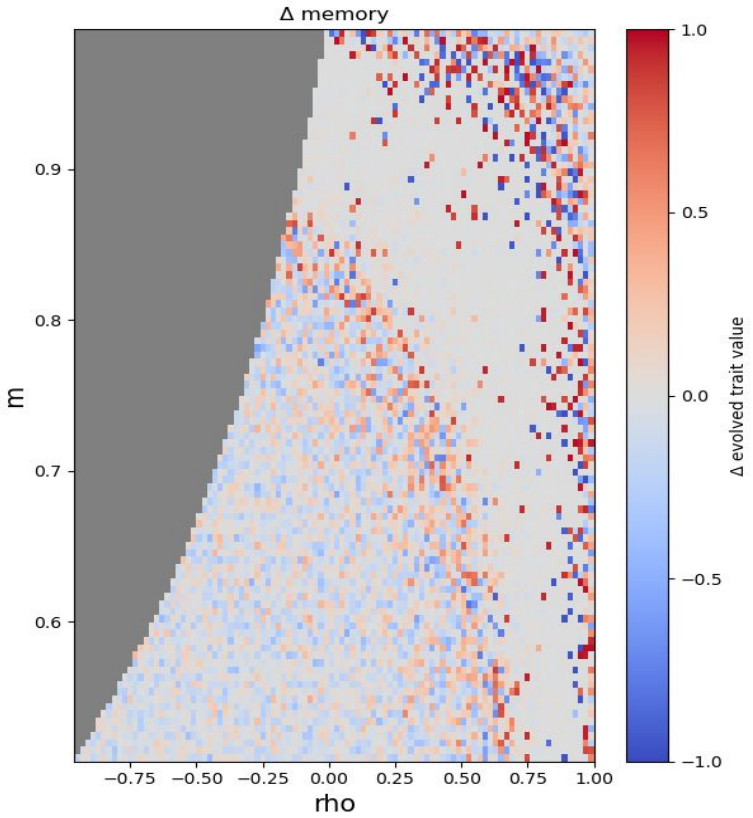


Strategy 2: Compare the Memory(dim ‘n’) - Memory(dim 1) and see how this ‘Δ’ changes with increasing dimensions



Here Δ= Higher dimension memory - 1D memory

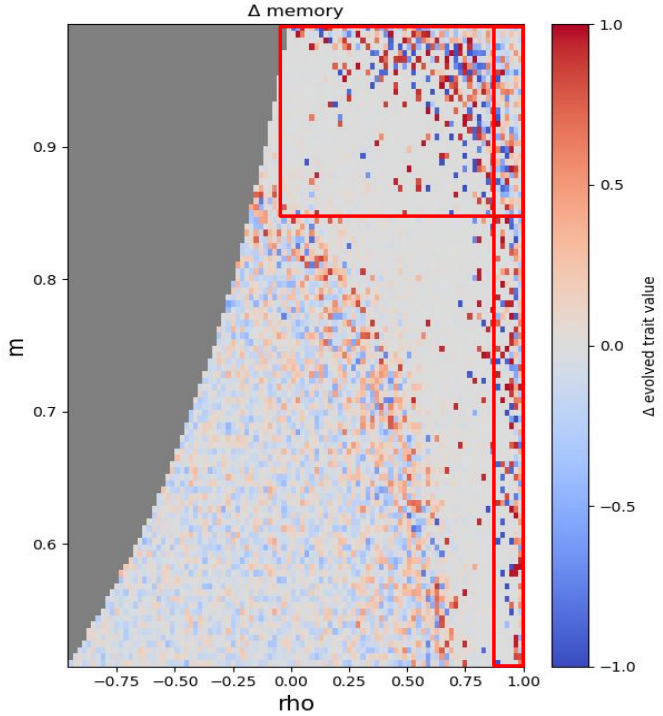
Strategy 2: Use delta plots; Compare the Memory(dim ‘n’) - Memory(dim 1) and see how this ‘delta changes with increasing dimensions



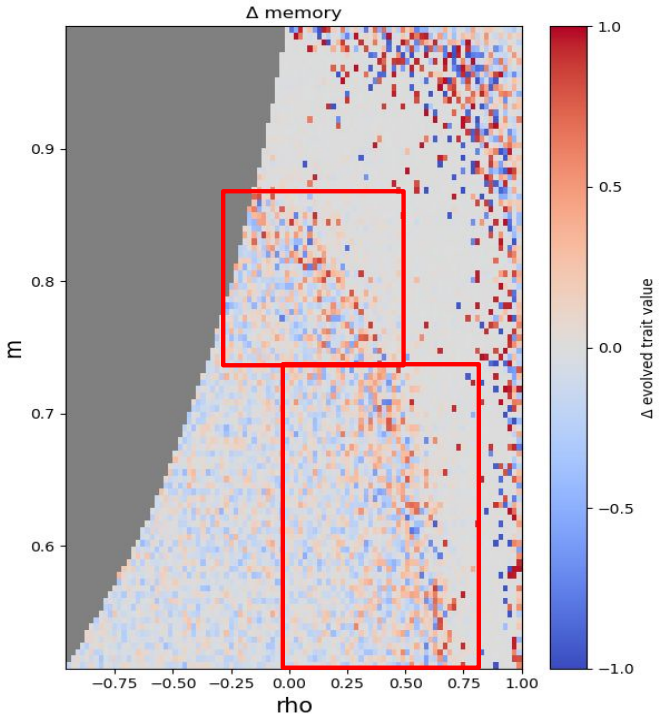
Positive Δ memory: digher dimension memory > 1 D memory

Negative Δ memory: higher dimension memory < 1 D memory

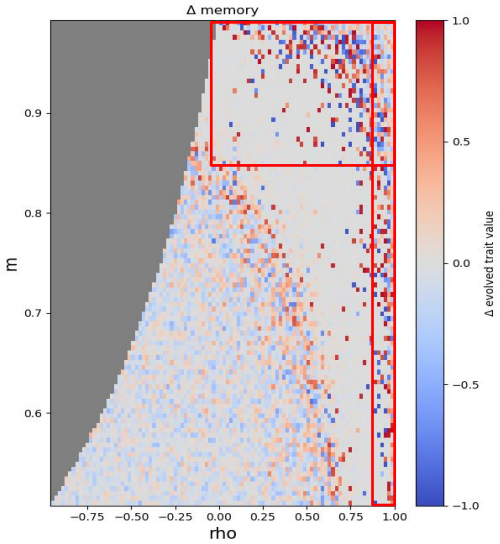
Strategy 2: Compare the Memory(dim ‘n’) - Memory(dim 1) and see how this ‘delta changes with increasing dimensions

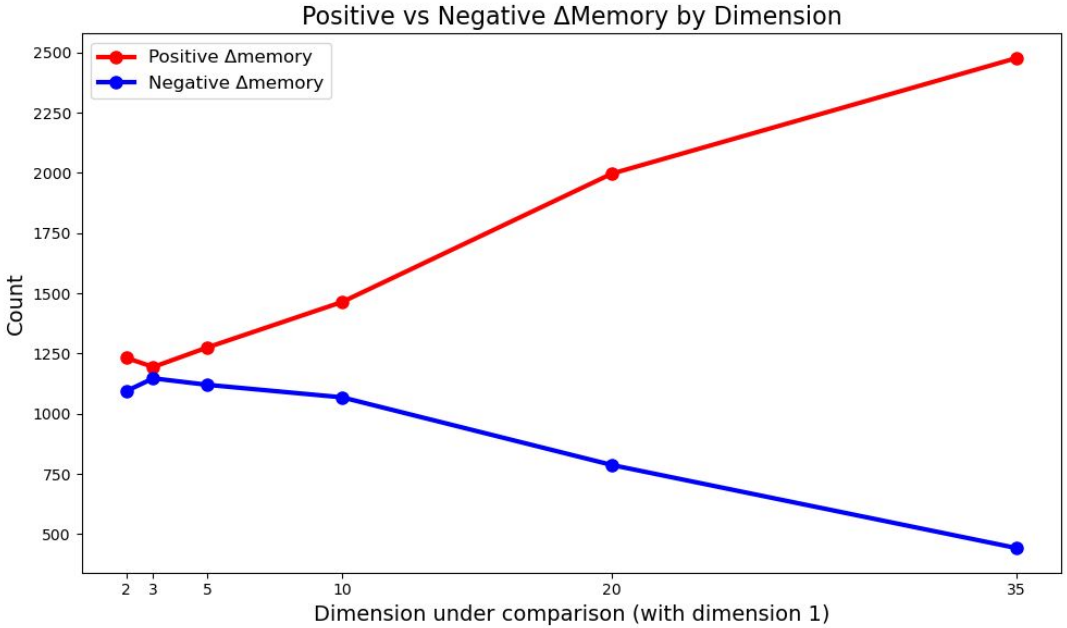


Zone 1: right corner

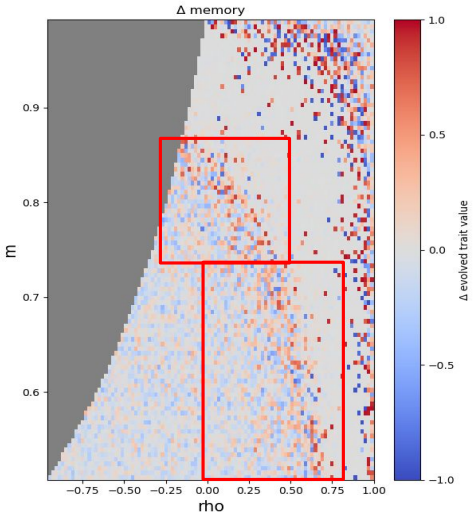


Zone 2: Center strip

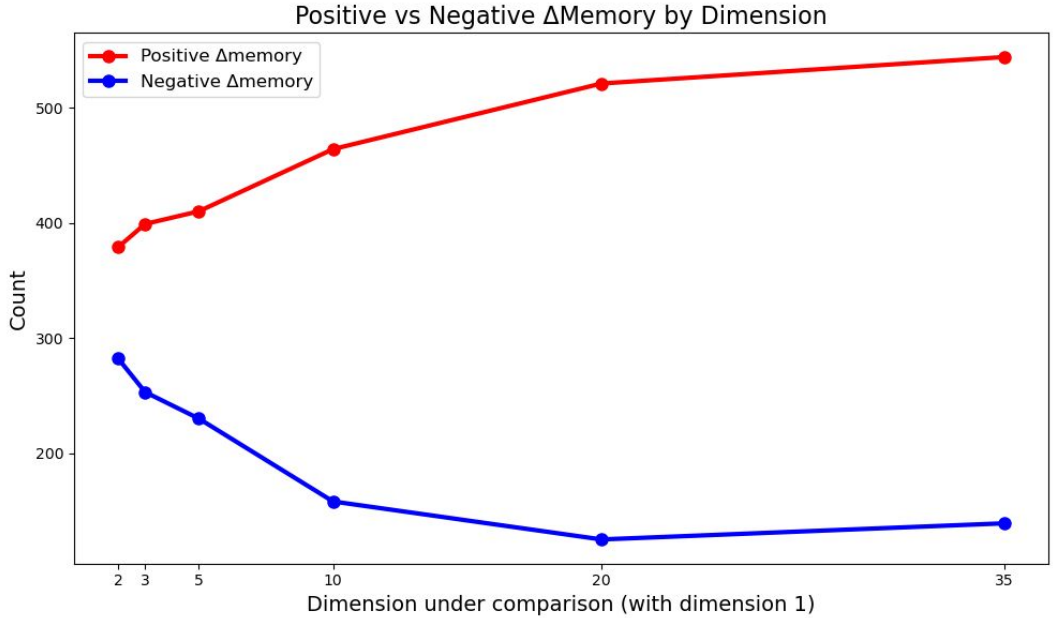




Here Δmemory = Higher dimension memory - 1D memory



Zone 2: Center strip



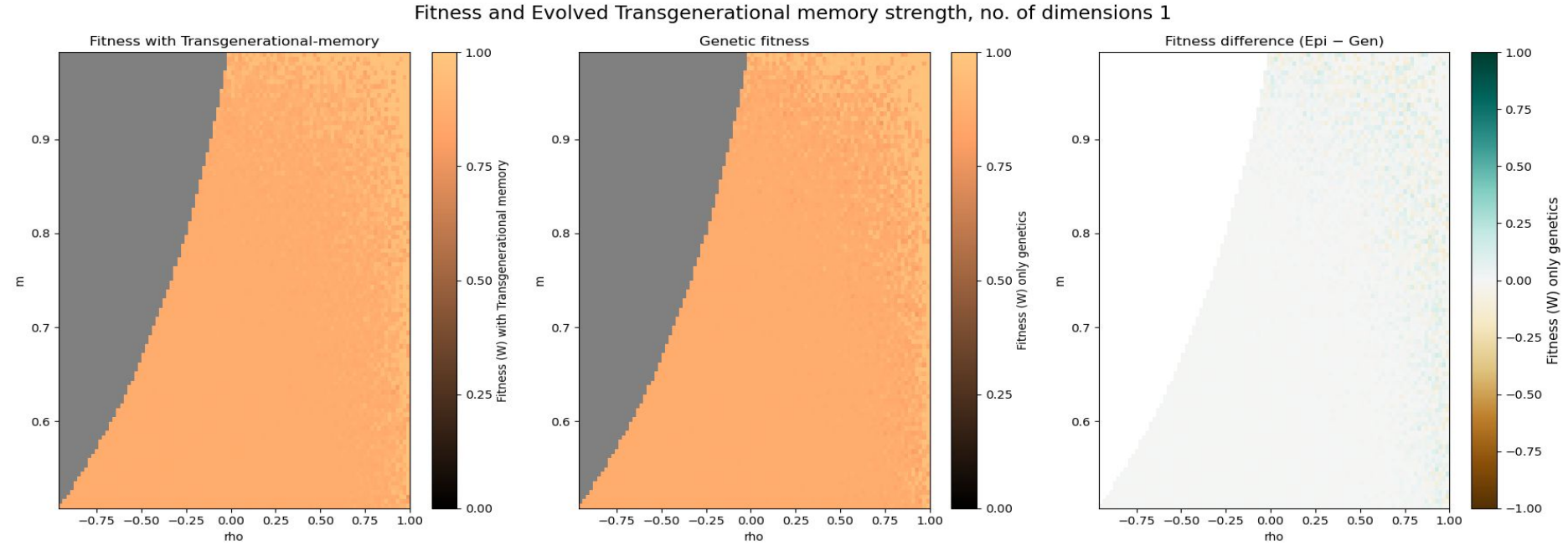
Here Δmemory = Higher dimension memory - 1D memory

Question 2: How does the presence of memory change the fitness of a lineage across dimensions?

Try and investigate this by subtracting the (epi-memory fitness − only-genetic) fitness compared across dimensions

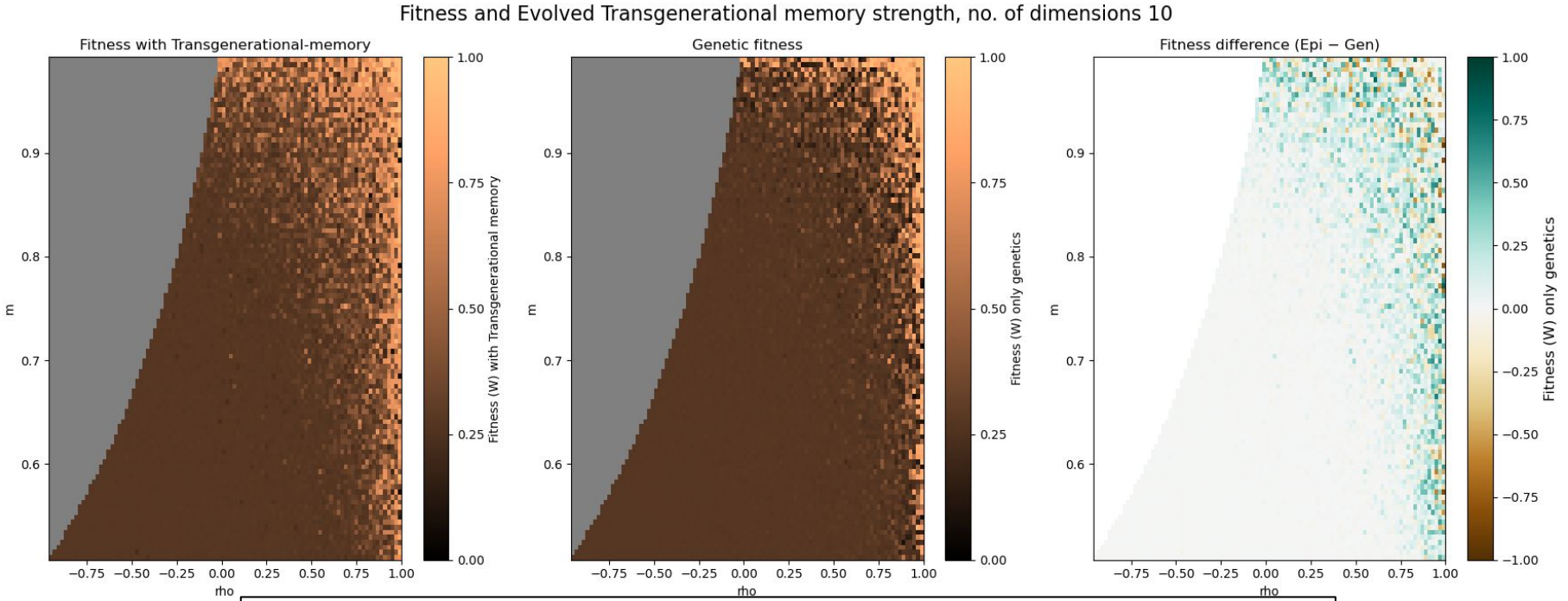
Here Δ Finess = Epi-fitness - Genetic fitness

Does Δ Finess = Epi-fitness - Genetic fitness change with increasing dimensions?



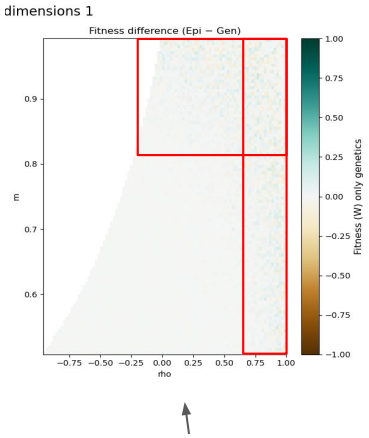
Here Δ Finess = Epi-fitness - Genetic fitness

Does Δ Finess = Epi-fitness - Genetic fitness increase or decrease with increasing dimensions

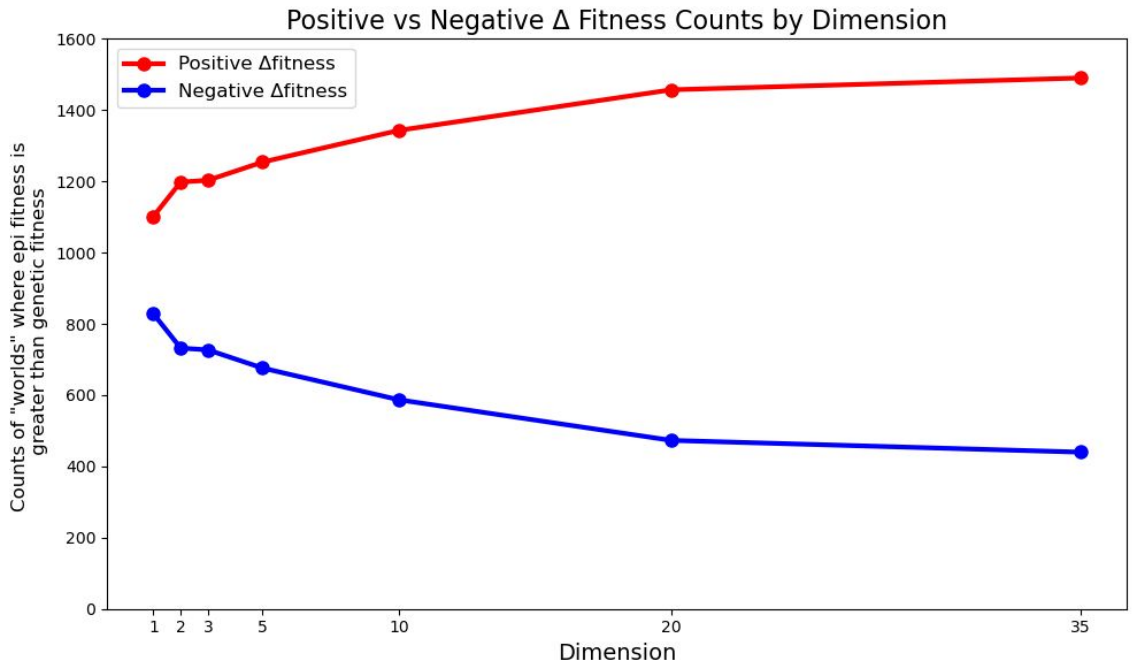


Here Δ Finess = Epi-fitness - Genetic fitness

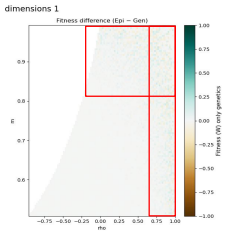
Does (Δ Finess = Epi-fitness - Genetic fitness) of the **right corner** increase or decrease with increasing dimensions

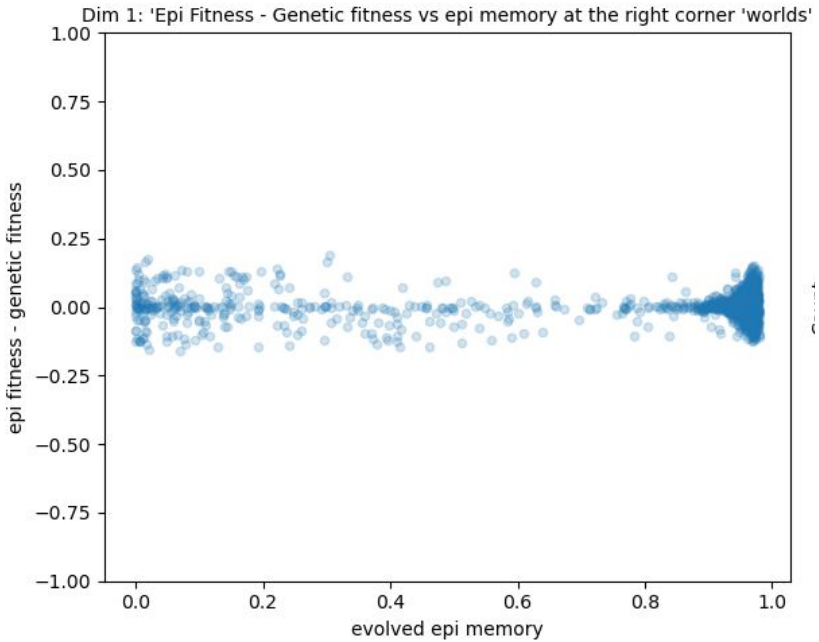


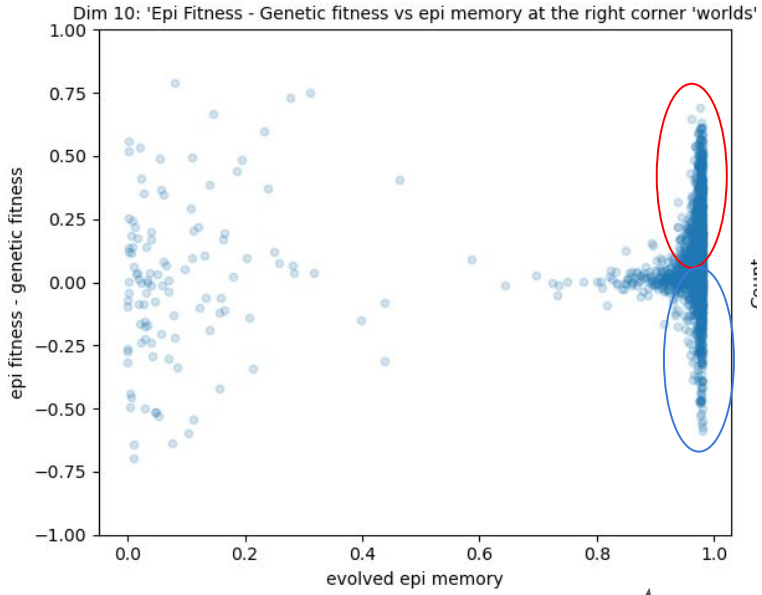
Right corner worlds: whats happening here? - Does the number of worlds with positive Δ Fitness (= Epi-fitness - Genetic) change with increasing fitness?



Then I examine how the Δ Finess = Epi-fitness - Genetic correlates with the value of evolved memory at the ‘right corner’

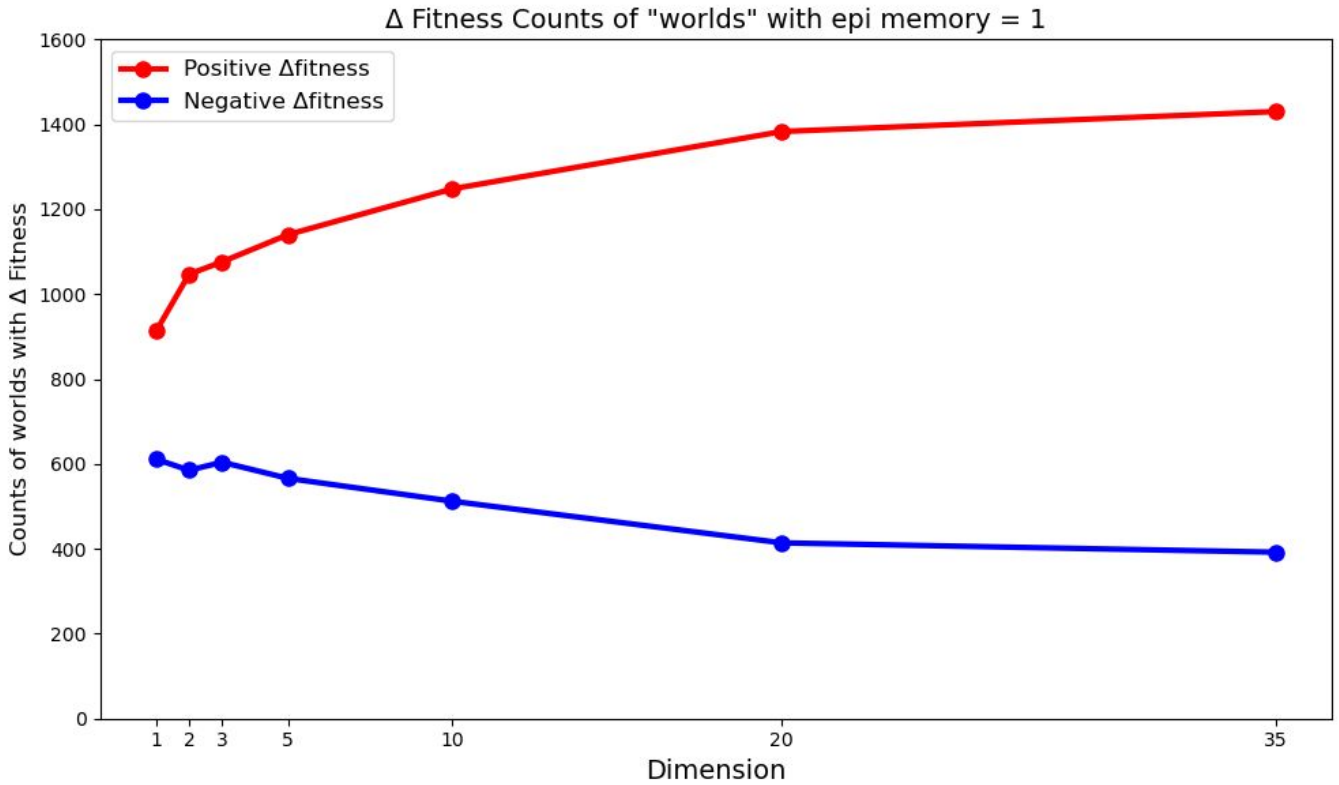






Both positive and negative Δ Fitness correlates with high (~ 1) epi memory in this region

Subsetting ‘worlds’ in the right corner with Epi memory = 1



But when added up, on average high memory tends to have positive Δ Fitness than negative Δ Fitness, and this effect increases with increasing dimensions

Results summary

1. Higher dimensional phenotypes expands the number of environmental parameter combinations ("worlds") that evolve positive transgenerational memory as well as the value of evolved memory in these worlds.
2. In the high-ρ, high-m parameter region, memory evolution outperforms genetics-alone strategies across more worlds. This advantage grows with phenotypic dimensionality, as higher dimensions expand the number of worlds where memory yields higher fitness.
3. High memory lineages (memory ~ 1) in the high-ρ, high-m region on average tend to have higher fitness than their genetic counterparts; and this effect amplifies with increasing dimensions
    a. High memory lineages in the rest of the parameter space show no increase in fitness in the presence of memory

1. Memory + dimensionality = adaptive in stable environments
2. Memory+ diemnsionality = still evolves in unstable environments: but why? – maybe explanations involving positive frequency dependence requires invoking